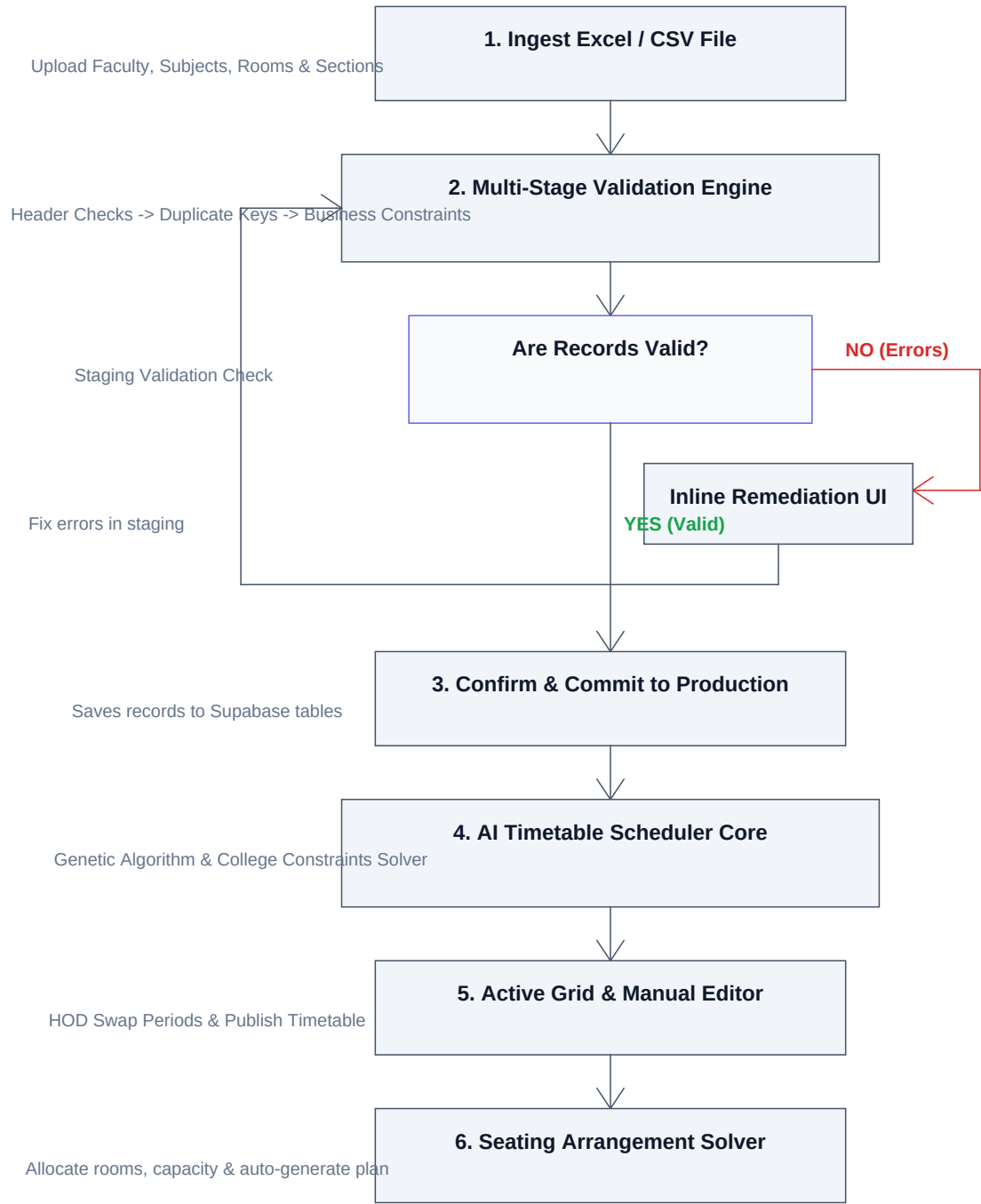


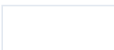
# AcadOps - AI Academic Timetable & Seating System

Complete System Operations Flowchart & Architecture Blueprint

## SYSTEM OPERATIONS & INGESTION FLOWCHART



### LEGEND:



Standard Operation Stage



Validation Decision check

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## DETAILED OPERATIONS & ARCHITECTURE DESCRIPTION

### 1. Data Ingestion Layer (Excel/CSV Parser)

HODs/Admins upload department data files (.csv, .xlsx, or .xls). The parser converts raw sheets (such as visual grids) into a unified, flat relational structure containing Department details, Academic Years, Sections, Subject Codes/Names, and Faculty mappings. (Pandas & OpenPyXL-powered back-end parsing).

### 2. Multi-Stage Validation Engine

The system runs parallel validations before inserting any records to prevent database corruption:

- Header Validation: Ensures exact column keys are present.
- Row Integrity: Checks for missing fields (e.g., empty emails or subject types).
- Duplicate Constraints: Blocks multiple conflicting listings for same subjects or duplicate user profiles.
- Business Rules Check: Flags anomalous assignments such as HOD teaching hours exceeding limits or incorrect lab durations.

### 3. Validation Staging & Inline Remediation

Rows with missing fields or warnings are placed in a staging state (yellow/red warnings). HODs do not need to re-upload files; they can correct warnings directly inside the UI preview. Once all records are marked VALID, they can commit the clean dataset to the production database.

### 4. AI Timetable Solver Core (FastAPI & Genetic Solver)

Applies mathematical heuristics to generate conflict-free timetables under strict constraints:

- HOD constraints: The Head of Department cannot be assigned to first or last periods, and has no afternoon periods on Wednesdays.
- Lunch breaks: Year 1 has lunch after period 3; Years 2, 3, 4 have lunch after period 4.
- Faculty limits: Max 2 consecutive teaching periods. Labs restricted to 2 or 3 periods based on branch configurations.

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## DATABASE ARCHITECTURE & EXAM SEATING WORKFLOW

### 5. Supabase SQL Storage Layer

A secure PostgreSQL layer hosted via Supabase. It uses connection pooling (Port 6543) and utilizes SQLAlchemy's 'connect\_args={"statement\_cache\_size": 0}' configuration in FastAPI to prevent transaction cache clashes. Includes tables for Users, Profiles, Subject Registry, Section Configs, Classrooms, and Timetable Slots.

### 6. Active Timetable Swap & Manual Fine-Tuning

Once a timetable is published, HODs can swap conflicting periods interactively via the frontend grid. The backend validates swap requests on the fly (preventing double-bookings of rooms or teachers) and updates the Postgres database instantly.

### 7. Auto Seating Arrangement Solver

The Seating Module loads registered students and classroom capacities (from the Classrooms table), calculates physical spacing constraints, and generates an optimized seat allocation grid. It prevents student double-allocations and structures room layouts dynamically.